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Studierichting: Informatica
Oefeningenreeks 4A nr. 26.11

$$\begin{aligned} & \int \frac{x^2 - 2x + 4}{x^2 - 2x + 1} dx \\ &= \int \frac{x^2 - 2x + 1}{x^2 - 2x + 1} dx + \int \frac{3}{x^2 - 2x + 1} dx \\ &= \int dx + 3 \int \frac{1}{x^2 - 2x + 1} dx \\ &= \left(x + 3 \cdot \int \frac{1}{x^2 - 2x + 1} \right) dx \\ &= \left(x + 3 \cdot \int \frac{1}{(x - 1)^2} \right) dx \end{aligned}$$

Stel $t = x - 1$ dan $dx = dt$

$$\begin{aligned} &= \left(x + 3 \int \frac{1}{t^2} \right) dt = \left(x + 3 \int t^{-2} \right) dt \\ &= x + 3 \left(\frac{-1}{t} \right) + c \\ &= x - \frac{3}{x - 1} + c \end{aligned}$$

References